

Exercise 1

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HW 1

$$(a) \quad d_m(x, z) = \sqrt{\sum_{i=1}^n (x_i - z_i)^T \underbrace{M}_{\text{scaling factor same as } (\hat{x} - \hat{z})} (x_i - z_i)}$$

$$= \sqrt{\sum_{i=1}^n (x_i - z_i)^T (\hat{x} - \hat{z})}$$

$$= \sqrt{\sum_{i=1}^n (x_i^T - z_i^T) (\hat{x} - \hat{z})}$$

$$= \sqrt{\sum_{i=1}^n (x_i^T \hat{x} - x_i^T \hat{z} + z_i^T \hat{z})}$$

because $\hat{x}$ and $\hat{z}$ are
1-dimensional
vectors

$$= \sqrt{\sum_{i=1}^n (x_i^2 m_i - x_i z_i m_i + z_i^2 m_i)}$$

$$= \sqrt{\sum_{i=1}^n (m_i x_i - m_i z_i)^2} = \sqrt{\sum_{i=1}^n (\hat{x}_i - \hat{z}_i)^2} = \|\hat{x} - \hat{z}\|_2$$

$$\equiv d(\hat{x}, \hat{z}) = \|\hat{x} - \hat{z}\|_2$$

(b) No